

LectureAssist: An Agentic Multimodal AI Assistant that Turns Lectures into Adaptive Study Material

Fully local, zero-paid-API question generation from lecture video and course documents

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ABSTRACT

Lecture recordings and course documents are everywhere, but the tools that turn them into real practice are not. Most study aids expect clean text pasted in by hand, or run on paid cloud APIs that few students can afford. LectureAssist takes a raw lecture video or a document and produces study and assessment material on its own, with no paid service anywhere. It reads the board and the slides with OCR, listens to the audio with Whisper, and then a team of agents plans, retrieves, writes, checks and rewrites quiz questions. Chat and adaptive difficulty close the loop.

Method with System Diagram/Design Complexity

An orchestrator coordinates specialist agents. Video is sampled into frames and audio demuxed; documents are parsed directly. Each frame is classified by brightness and edge statistics as slide, blackboard, whiteboard or scene, then preprocessed before OCR; Faster-Whisper transcribes speech. Both channels fuse into one timestamped timeline, then summarised, chunked, embedded and indexed in FAISS. A Quiz Planner allocates topic-difficulty slots; a Generator drafts one question at a time, invoking retrieval as a tool when context is insufficient; a Validator returns PASS or FAIL with a remediation action; a Refiner retries within budget. Models are Gemma 3, served locally by Ollama.

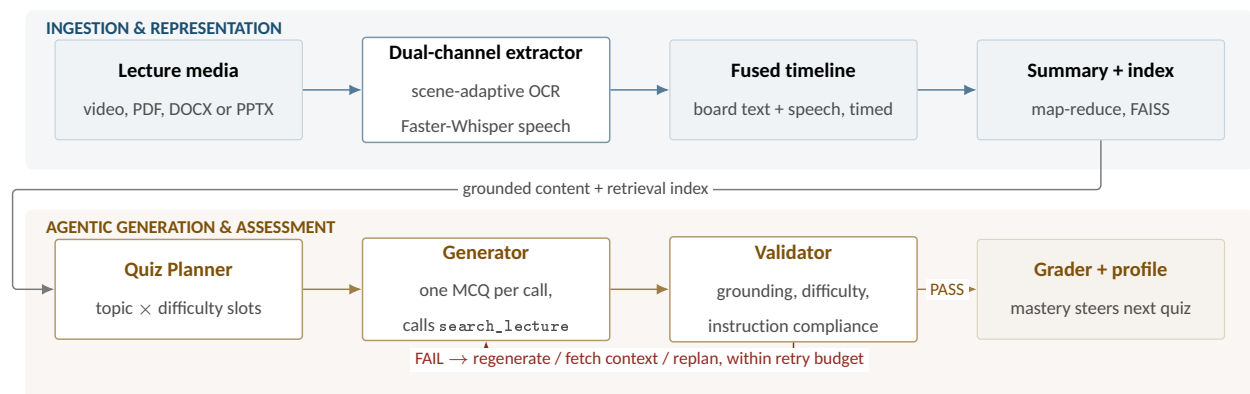


Figure 1: LectureAssist system architecture. Row 1 turns raw media into one grounded representation; row 2 is the agentic loop, where a failed question is diagnosed and routed back through the Refiner, not discarded.

Novelty of Project

Two elements are new. Extraction is scene-adaptive rather than a single OCR recipe, so handwritten board work and projected slides survive. And the validator diagnoses a failed question, choosing whether to regenerate, fetch context or replan the topic, instead of merely filtering. Comparable tools need pasted text or paid APIs.

Impact on Society/Environment

Retrieval practice is a highly effective study strategy, but good practice questions are unevenly distributed. Generating them from a student's own lectures, on hardware they already own, removes a cost barrier rather than an ability barrier, supporting SDG 4 and 10. Local 4B inference also avoids datacentre-scale energy per query.

Business Model/Feasibility/Financial Scalability Plan

Marginal cost per student is near zero: no paid API is used, and the only input is compute the user already

owns. The plan is a free, open-source self-hosted tier for individual students, and a paid institutional tier in which a department runs one GPU server for an entire course, licensed per seat per semester at a fraction of per-token cloud pricing. Revenue comes from institutional licences, LMS integration and support rather than student subscriptions. Feasibility is already demonstrated: the whole system runs on a free-tier T4, so one modest GPU can serve a class.

Conclusion

The pipeline, extraction, retrieval, generation, grading and adaptation, is implemented and running. Two of four generation pipelines have completed a scored run of 200 MCQs each; on native metrics the non-agentic baseline currently matches or beats the agentic loop, which duplicates more. EQGBench and ReQUESTA harnesses are built but unrun.